Decodable but Not Load-Bearing: Rethinking Physics Injection in Latent World Models

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Abstract

A world model predicts how an environment evolves in response to actions by rolling an internal representation forward, and is expected to respect physical regularities such as how objects move and collide. We study JEPA-style latent world models and find a puzzle: trained with no physics supervision, such a model already recovers object position well from its representation and predicts one step almost exactly, yet its multi-step rollouts drift away from the very dynamics it encodes—the physical state is *decodable, but not load-bearing*. The popular fix is to inject physical structure into the latent representation. We test this exhaustively—ten injection variants spanning hard and soft constraints on both physical quantities and physical laws, with labeled and label-free supervision—across three controlled physics domains and two realistic simulation benchmarks, and find injection fails in 29 of 30 cases; even supplying the *correct* physical constraint makes prediction worse. We trace this to a single mechanism: the physical state is already encoded across the model's internal representation, so an injected copy is superfluous—prediction reads the state from the internal representation and bypasses the injected one. What the model lacks is not the object's physical state but the ability to evolve that state over long horizons: fixing the training procedure alone, adding no physics, cuts rollout error 2.2–8.3× on identical code and metrics. To help a latent world model, injected structure should supply what the representation is missing—*how the state evolves, not what the state is*.

1 Introduction

A *world model* rolls an internal representation forward to predict how an environment evolves under an agent's actions—a simulator to plan with instead of acting in the world (Ha and Schmidhuber 2018; LeCun 2022). For a physical environment, it should respect how objects move, fall, and collide. JEPA-style latent world models (Maes 2025; Balestriero and LeCun 2025) present a puzzle here. Trained on PhyWorld (Kang et al. 2025) with no physics supervision, such a model already recovers object position from its representation by linear probing (correlation up to 0.96) and predicts one step almost exactly (latent cosine 0.98–0.99)—yet rolled out autoregressively, its predictions drift away from the very dynamics it encodes: on collision the latent cosine falls to 0.24 by horizon 28, and out-of-distribution (OOD) physics—unseen radii, masses, or velocities—multiplies rollout error several-fold. This indicates the state is already present in the representation; the model simply fails to *evolve* it.

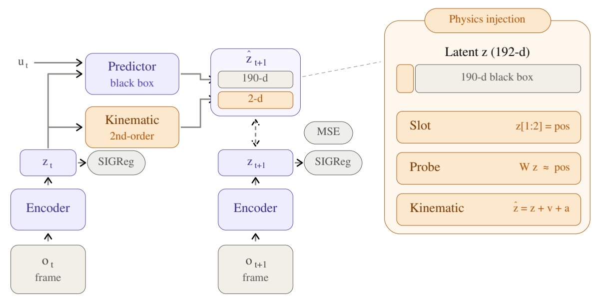

Figure 1: Physics injection into the LeWM shared-latent world model. *Amber* marks everything injection adds to the backbone; the rest is unmodified LeWM. The encoder maps each frame to a 192-d latent; a black-box transformer predicts the full latent from the latent history and action u_t, and the free-rollout objective aligns the predicted $\hat{z}_{t+1}$ to the encoded next latent by MSE (SIGReg prevents collapse). Physics enters a 2-d position *slot* carved from the 192 dimensions (right panel): a hard *slot* pinning $z[1{:}2]$ to position, a supervised *probe* reading position off the latent, or a *kinematic* head evolving the slot by a second-order rule—while the other 190 "black-box" dimensions stay free. This 2-vs-190 split is the crux: prediction can route through the black box and bypass any injected slot.

The popular fix is to *inject physical structure into the latent* (Figure 1): a dedicated slot pinned to the physical state, a supervised probe that reads it out, or a hard-wired dynamics rule. Such injection has real successes in adjacent model classes—architectures that route prediction through a low-dimensional physical state by construction (Mao, Umasudhan, and Ruchkin 2024; Peper et al. 2025), deep supervision where the supervised quantities occupy a large fraction of a compact state latent (Zahorodnii 2025), and a decomposed JEPA latent supervised on low-dimensional numerical time series (Nie et al. 2026). Scaling data is not the answer either: video generators trained at scale still fail OOD physics, memorizing cases rather than laws (Kang et al. 2025). But for a *shared, high-dimensional* latent—where prediction runs end-to-end through one representation—injection has only been tried in isolated, incomparable settings, and never against the confound that dominates the outcome: the training protocol. Whether physical structure adds anything, once the protocol is done right, has never been tested systematically.

We answer this with a systematic, controlled anatomy on a JEPA world model. We cross ten injection variants—spanning hard and soft constraints, targeting the physical *state* (what it is) or its *evolution* (how it changes), under labeled and label-free supervision—with two training regimes and three controlled physics domains, then corroborate on two realistic-simulation benchmarks (Bear et al. 2021; Tung et al. 2023).

Across the full grid, the structure does not help: injection degrades or fails to improve prediction in 29 of 30 mechanism–domain cells. Supplying the *correct* evolution rule—the gravitational law $a{=}g$ wired into the state update—does no better than a freely learned acceleration where $a{=}g$ *is* the ground truth (0.173 vs. 0.178 nMSE on parabola), and degrades uniform motion by $1.57\times$. Co-training the structure from scratch, rather than grafting it onto a pretrained encoder, does not rescue it either—injection costs accuracy in all six regime–domain cells, and from-scratch training yields the worst failure we observe ($+0.558$ nMSE on uniform, against $+0.035$ fine-tuned)—closing the defense that physics must be baked in during pretraining. The one cell that does improve reverses sign on the other two domains: using it would mean knowing each domain's dynamics in advance.

Why should the *correct* physics hurt? Because the physical state is *already there*. A pinned slot occupies 2 of the 192 latent dimensions, but the other 190 redundantly encode the same position—decoding it nearly as accurately as the full latent—so prediction routes around the injected slot and reads the copy it already carries. This account makes a falsifiable prediction: if the slot's $\sim$1% share of the prediction loss is what starves it, up-weighting the slot should remove the harm. It does not—re-weighting to saturation recovers only *parity*, never a net gain, because no loss weight closes a bypass wired into the architecture. The state is *decodable, but not load-bearing*—present in the latent, yet carrying none of the prediction.

What the representation lacks is not the state but the ability to *evolve* it over long horizons, and that gap is fixable with no physics at all. Repairing the training protocol alone does it: free rollout, with the horizon matched to each domain's dynamics, cuts rollout error 2.2–8.3$\times$ on the identical code, metrics, and seeds that judged every injection cell. That pins the failure on the injected structure rather than on the pipeline or the yardstick. The lesson is constructive: an inductive bias must supply what a shared latent is missing (how the state evolves), not re-encode what it already holds. Only an architecture can close a wired-in bypass. Porting one that makes the physical state the *sole* pathway for prediction (Mao, Umasudhan, and Ruchkin 2024) does close it, but it collapses once an OOD shift reaches its encoder (ρ 0.33 vs. 0.89): closing the bypass is necessary, but not sufficient. Every finding above replicates on a second, frozen-DINOv2 backbone.

This paper contributes:

- to our knowledge, the first systematic anatomy of physics injection in *shared-latent* visual world models—ten variants $\times$ two training regimes $\times$ three domains, plus two realistic-simulation benchmarks—finding injection no better than baseline in 29 of 30 cells, and replicating on a frozen-DINOv2 backbone (§4.2, §4.5);
- a mechanistic account, *decodable but not load-bearing*: the 190 black-box dimensions decode position as accurately as all 192, and re-weighting the slot to saturation buys only parity, so the bypass is architectural. A faithful extrinsic port confirms this from the outside: it closes the bypass, yet collapses under encoder-OOD shift (§4.3, §4.5);
- a controlled factorial showing the dominant variable is the training protocol, not physical structure: free rollout alone cuts rollout error 2.2–8.3$\times$ with no physics added, doubling as the positive control against implementation and metric artifacts (§4.4).

2 Related Work

Physically structured world models. A growing line argues that world models should carry explicit physical state: physically interpretable models route prediction through a dedicated low-dimensional state (Mao, Umasudhan, and Ruchkin 2024; Peper et al. 2025); deep supervision adds physical readouts to the loss and helps in a low-dimensional state latent, where the supervised quantities occupy $\sim$38% of the representation (Zahorodnii 2025); closest to ours, Phys-JEPA supervises a decomposed JEPA latent and reports gains (Nie et al. 2026)—but on numerical time series, not the visual latents we study, a distinction that reverses the conclusion (§4.3). Physics-informed losses (Raissi, Perdikaris, and Karniadakis 2019) and conservation-structured dynamics (Greydanus, Dzamba, and Yosinski 2019; Cranmer et al. 2020) constrain the model class; graph simulators over *explicit* particle state predict physics well (Battaglia et al. 2016; Sanchez-Gonzalez et al. 2020); architectural biases such as restricted attention let transformers recover force laws from numerical trajectories (Liu et al. 2026)—structure works when the physical state *is* the representation, not a small part of a shared one. We do not dispute these results in their original settings; we test the transferable core of these proposals inside a shared-latent *visual* JEPA model and find all of them unhelpful or harmful (§4.2); our extrinsic port (§4.5) attributes the gap to architecture, not to the physics.

Latent world models and physics benchmarks. JEPA-style models predict in representation space (Assran et al. 2023; Bardes et al. 2024; Balestriero and LeCun 2025; Maes 2025); object-level masking structures the JEPA objective itself rather than the latent's physical content (Nam et al. 2026); DINO-WM plans over frozen features (Zhou et al. 2025; Oquab et al. 2024), the design we adopt as our cross-backbone control; Dreamer-line models train on imagined rollouts (Ha and Schmidhuber 2018; Hafner et al. 2025). PhyWorld (Kang et al. 2025) supplies controlled single-mechanism physics videos with explicit in-distribution ranges, showing that scaled video generators memorize cases rather than laws; Physion and Physion++ (Bear et al. 2021; Tung et al. 2023) add realistic—not camera-captured—simulation. We apply PhyWorld's OOD protocol to *latent* models rather than pixel-space generators.

Rollout training and exposure bias. Training predictors on their own outputs is a classic remedy for compounding error (Bengio et al. 2015; Venkatraman, Hebert, and Bagnell 2015; Goyal et al. 2016). We claim no novelty for free rollout; its role here is methodological (§3.4): the controlled factorial variable prior injection studies never controlled, and a positive control separating "structure does not help" from "the implementation or metrics are broken."

3 Method: A Design Space of Physical Injection

We answer the question of §1 by construction: design the ways physics can be injected into a shared latent, state what each design is supposed to achieve, and lay out the experiments that test the mechanism behind it.

3.1 Setting and Notation

We study LeWM (Maes 2025), a compact JEPA-style action-conditioned world model ($\sim$10M parameters): an encoder E_θ (ViT-tiny) maps frame o_t to a latent $z_t \in \mathbb{R}^D$ with D=192, and a causal-transformer predictor F_ϕ rolls latents forward from the history and action u_t, i.e. $\hat{z}_{t+1} = F_\phi(z_{\le t}, u_t)$. Training minimizes a prediction loss, an anti-collapse regularizer (Balestriero and LeCun 2025), and—when physics is injected—an auxiliary physical loss:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \beta\,\text{SIGReg}(z) + \lambda\,\mathcal{L}_{\text{phys}}. \tag{1}$$

Here $\mathcal{L}_{\text{pred}} = \frac{1}{D}\sum_i w_i(\hat{z}_{t+1,i} - z_{t+1,i})^2$ is a per-dimension weighted prediction error. By default w_i=1; *re-weighting* raises w_i to w on the 2 slot dimensions (w swept $1 \to 300$ in §3.3), while λ scales the soft auxiliary losses (probe, consistency, label-free priors). Throughout, w denotes the slot's prediction-loss weight and λ an auxiliary-loss weight—two distinct dials. *Teacher forcing* trains F_ϕ on one-step transitions from encoded ground-truth context; *free rollout* feeds predictions back for H autoregressive steps (H=8 unless stated) and supervises the whole rollout. The kinematic head (Group 2 below) replaces the slot's transition by a second-order update

$$\hat{z}_{t+1}^{[0:2]} = z_t^{[0:2]} + v_t\,\Delta t + \tfrac{1}{2}\,a\,\Delta t^2, \tag{2}$$

where v_t is the finite-difference slot velocity and the acceleration a is either a learned $\text{MLP}(z_t, u_t)$ or a single learnable constant—the strict a=g head. Unless stated otherwise the encoder is initialized from the public PushT checkpoint; §4.2 additionally studies from-scratch training.

Variant	What it does, and what it tests
Group 1: inject the state — what the motion *is* (position, velocity)	
slot (structpos)	*hard.* pin dims $[0{:}2]$ to the true position (the object's x, y coordinates; a fixed index range, not an attention slot (Locatello et al. 2020)): the most direct state injection
+reweight	*hard.* raise the slot's share of the prediction loss (w=30, Eq. 1): if failure is due to the slot's $\sim$1% gradient share, this should rescue it
+velocity	*hard.* encode velocity too: tests whether it matters *which* quantity is injected, not just that something is
probe	*soft.* linear head reads position from the latent: replicates the deep-supervision recipe (Zahorodnii 2025) in a high-dimensional visual latent
+slot	*both.* soft + hard combination: tests complementarity
Group 2: inject the evolution — how it *changes* (a dynamics rule)	
free MLP	*hard.* attach z+v+a update with learned a: upgrade from pinning state to pinning evolution
strict a=g	*hard.* exact gravitational form, one learnable constant: closes the "your physics was wrong" objection
consistency	*soft.* finite-difference velocity of the rolled-out positions must match ground truth; assumes no form for a, designed for collision impulses
Group 3: drop the labels — is ground truth *needed*? (plus a control)	
label-free	*soft.* slot must evolve as smooth second-order dynamics, no ground truth: the only injection available for raw video
grounded	*soft.* same structure + true labels: isolates the label variable

Table 1: Ten injection variants in three groups. The groups follow the **target** axis—inject the *state* (what the motion is), its *evolution* (how it changes), or, dropping **supervision**, a label-free prior over that evolution with its labeled control. The **constraint** axis is marked per variant: *hard* forces a latent dimension to equal the physical quantity, *soft* adds an auxiliary loss that only encourages it. These three axes span the ways to inject physics into a shared latent, so any alternative encoding falls on an axis we test. All variants use the strongest training protocol (§3.4).

3.2 Injection Design Space: Three Groups, Ten Variants

We span the design space along three orthogonal axes—injecting *what the motion state is* vs. *how it evolves*, under *hard vs. soft* constraints, with *labeled vs. label-free* supervision—so that any "try a different encoding" objection lands on an axis we have measured. Table 1 groups the ten variants by the first axis and gives the design rationale of each.

Orthogonally, we cross injection with the *training regime*: fine-tuning a pretrained encoder vs. from-scratch co-training; with injection on/off this gives a 2×2 per domain.

This closes the remaining escape hatch—"structure must be baked in during pretraining, as extrinsic architectures do."

3.3 Mechanistic Hypothesis and Its Tests

Hypothesis (the load-bearing problem). The physical slot occupies 2 of 192 dimensions and receives $\sim 1\%$ of the prediction gradient, while the black-box channel *redundantly* encodes the same position—so prediction can route around any injected slot. Because the latent already carries the injected state, injection adds no new signal—it only consumes capacity and gradient share. We test this account in three ways, each admitting a specific outcome that would falsify it:

1. **Is the physical constraint injected?** Split the latent into the 2 slot dimensions and the 190 black-box dimensions and decode position from each separately, with a random-2-dimension control (Hewitt and Liang 2019). If the black box alone cannot decode position, the redundancy claim is false.
2. **Is the constraint acting at all?** Two measurements: how large the weighted physical auxiliary loss is relative to the prediction loss (a proxy for gradient magnitude), and whether the encoder's effective dimensionality (participation ratio) changes. If neither moves, the constraint never took effect and the failure is a bug. If the auxiliary term dominates and dimensionality collapses, the constraint is acting—and competing with prediction.
3. **Does re-weighting the physical loss help?** Sweep the slot weight w of Eq. 1 from 1 to 300. If harm does not respond systematically to the weight, the gradient-share mechanism is wrong; if it responds but weighting to saturation still cannot rescue every domain, demonstrating that insufficient physical loss is not the root cause.

3.4 The Training-Protocol Control: Free Rollout

Free rollout serves two methodological roles; neither is a novelty claim (it is scheduled sampling, Bengio et al. 2015).

Controlled factorial variable. Physical structure has never been compared against the training protocol on the same baseline; prior studies treat the protocol as background. We set the protocol first—replacing teacher forcing with free rollout (§3.1)—and then ask whether structure adds incremental value. Teacher forcing hides compounding error at training time (*exposure bias*): the model never sees its own mistakes, then must consume them at deployment. Free rollout exposes them during training, injecting no physics whatsoever. We read the resulting gain as more than robustness to compounding error: supervising an H-step autoregressive rollout turns training itself into *long-horizon dynamics simulation*. Under teacher forcing the predictor is graded only on reproducing local one-step transitions; under free rollout it is graded on *evolving* the state over the full horizon, so the gradient rewards a self-consistent multi-step dynamics rather than a one-step map—precisely the capability the latent is missing (§1). On this reading, free rollout is an evolution-targeted training signal that requires no physics labels: it trains *how the state evolves* without touching *what the state is*. A second protocol variable, the rollout horizon H, must *match dynamics complexity*: domains with late causal events (collision; realistic-simulation scenes) need horizons long enough to contain them.

Positive control. A negative result invites two deflations—"your implementation is broken" or "your metrics cannot detect improvement." A protocol change that produces large gains on identical code and identical metrics rules out both at once, pinning the failure on the injected structure itself.

3.5 External and Cross-Backbone Controls

Two controls probe our account from outside the LeWM backbone: one tests whether an architecture that *closes* the bypass succeeds where injection fails, the other whether the conclusions survive a different encoder.

Extrinsic port. We faithfully port the physically interpretable world model (PIWM) of Mao, Umasudhan, and Ruchkin (2024)—a three-stage pipeline (VAE encoder, 128-D; a supervised extractor to a low-dimensional physical state; known-equation dynamics with learnable constants)—onto the same PhyWorld domains and OOD protocol. The port tests our mechanism's architectural prediction from the outside: extrinsic designs make the physical state the sole pathway prediction must route through, closing the bypass by construction.

Cross-backbone control. To test that the conclusions are not artifacts of one encoder, we replicate the core protocol on a second JEPA-family instantiation in the style of DINO-WM (Zhou et al. 2025): a *frozen* DINOv2-small encoder (Oquab et al. 2024)—generic self-supervised features that have never seen PhyWorld—with the trainable projector serving as adapter and the identical predictor stack, losses, seeds, and evaluation. The variant differs from LeWM in backbone provenance, dimensionality, and, crucially, in whether the injected losses can reshape the encoder at all.

4 Experiments

4.1 Setup

Model. The compact, fully controlled backbone of §3.1 (LeWM: E_θ, an MLP projector, an action encoder, and a 6-layer causal-transformer F_ϕ) allows the complete factorial anatomy ($\sim$200 training runs), with headline configurations repeated over three seeds (3072/1234/42); remaining cells are single-seed and marked as such.

Datasets and OOD protocol. All benchmarks are *visual* world-modeling tasks (RGB video frames), not the numerical/tabular sequences of prior physics-injection work (§2); injected state is thus always a small fraction of a high-dimensional visual latent. *PhyWorld* (Kang et al. 2025) provides three single-mechanism synthetic domains—*uniform motion*, *parabola*, *collision* (zero/constant/impulsive acceleration)—each with 1,000 ID trajectories (32 frames, 224×224; radius/mass $\in [0.7, 1.5]$, velocity $\in [1, 4]$), evaluated on four partitions (ID, r/m-OOD, v-OOD, both-OOD); its single-variable OOD ranges isolate physical extrapolation from data-scale confounds. Two *realistic-simulation* benchmarks (rendered, not camera video) add realism: *Physion* (Bear et al. 2021) for

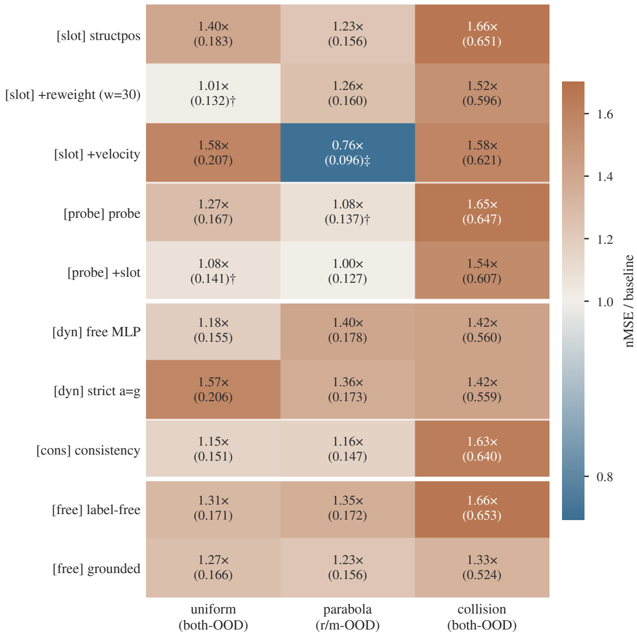

Figure 2: Every injection variant × domain against the free-rollout baseline (nMSE ratio; † = three-seed mean and ‡ = three-seed mean that beats baseline, others single-seed). Warm = worse, pale = parity, blue = better; heavy rules separate the three groups of Table 1, light rules the mechanisms within a group. Injection is at or below baseline in 29 of 30 cells (25 worse, 4 parity); the collision column (1.33–1.66×) is uniformly worst; the sole real gain is velocity+slot on parabola (0.76×, ‡).

zero-shot object-contact-prediction (OCP) transfer, and *Physion++* (Tung et al. 2023)—ten physical-property scenarios (mass/friction/bounce/deformation)—for direct training with rollout to horizon 64.

Metrics and decision rules. Verdicts use two scale-aware, *external* metrics—latent nMSE ($\|\hat{\mathbf{z}} - \mathbf{z}\|^2$ over target variance) and pixel PSNR of decoded rollouts—external because no injection variant optimizes them directly, and they agree in every case we examined. Latent cosine and linear-probe decodability ρ (Alain and Bengio 2016) are *diagnostics only*: both are duals of the auxiliary losses and rise mechanically when those are optimized, overstating physical competence (we measured cosine $1.50\times$ better while nMSE degraded to $0.21\times$). nMSE fails oppositely—its denominator degenerates when target variance collapses—so parabola is judged on its clean r/m-OOD partition, and all numbers are reported per partition and per horizon. Full pathology catalogue: Appendix E. Splits are trajectory-level; three-seed results report mean±sample std.

4.2 Injection Fails in 29 of 30 Cells

Injection degrades or fails to improve prediction in 29 of 30 mechanism–domain cells (25 worse, 4 at parity; Figure 2). Every cell carries the raw nMSE alongside the ratio, so the figure doubles as the full result table (seed annotations in Appendix B); read group by group, it answers every design rationale of Table 1. *Hard state injection*: the slot faithfully absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$; slot decodability rises $0.31 \rightarrow 0.96$), yet prediction worsens in all three domains ($1.40\times$/$1.23\times$/$1.66\times$)—the constraint acts, harmfully. *Soft injection*: the deep-supervision recipe, effective in a low-dimensional state latent where supervised quantities occupy $\sim$38% of the representation (Zahorodnii 2025), does not transfer to a 192-D visual latent where they occupy 2%: all three domains degrade ($1.27\times$/parity-after-seeds/$1.65\times$), and combining soft with hard does not rescue ($1.54\times$ on collision). *Correct physics does not help*: the strict $a{=}g$ head—the ground-truth functional form—degrades uniform by $1.57\times$ and parabola from 0.127 to 0.173, no better than an unconstrained MLP (0.178). *The purpose-built design fails worst on its target*: the consistency loss, designed for collision impulses, is among the worst exactly there ($1.63\times$). *Labels are not the issue*: the label-free prior and its perfectly labeled counterpart fail in the same direction (collision $1.66\times$ vs. $1.33\times$)—the failure is architectural, not informational.

The sole exception is the mechanism's signature, not a usable prior: adding *velocity* to the re-weighted slot helps on parabola (r/m-OOD $0.122{\pm}0.007 \rightarrow 0.096{\pm}0.007$, every seed lower) because there velocity is an extrapolable driving variable ($a{=}g$ makes it linear in time); the same encoding is the *worst* variant on uniform ($+0.076$, velocity is a redundant constant) and clearly harmful on collision ($+0.025$ over the reweighted slot; velocity jumps discontinuously). A prior that requires knowing the domain's dynamics in advance, and reverses sign across domains, is a diagnostic, not a method—and its best case (-0.026 nMSE) is an order of magnitude below the protocol lever of §4.4.

From-scratch co-training does not rescue injection. The remaining defense—structure must be baked in during pretraining—does not hold (Table 2): all six regime–domain cells are positive, so injection costs prediction accuracy whether the structure is co-trained from random initialization or grafted onto a pretrained encoder. Baking it in from the start also produces the single worst failure in the grid: on uniform the penalty is $+0.558$ ($3.9\times$) from scratch against $+0.035$ ($1.27\times$) fine-tuned. The two regimes trade places on the other two domains, so we do not read a general ordering between them—the robust statement is that neither escapes the harm.

Two corroborations independent of PhyWorld nMSE. First, *zero-shot transfer* (different dataset, metric, and task): on Physion OCP, no synthetic-trained configuration exceeds the random-encoder prior (0.607 mean AUC over the eight scenarios); the physics-structured variant is the *worst* of all (0.551), below free rollout (0.603), which approaches the prior monotonically over training (0.554 at epoch 1 to 0.603 at epoch 20) without ever crossing it. Per scenario, only Support ($+0.10$) and Collide ($+0.07$) clear the random control at all—"more structure, worse transfer," and most of the apparent zero-shot competence is the architecture, not the learning (Appendix E). Second, *realistic simulation*: trained directly on Physion++, every injection variant we ported

Domain	Regime	phys. off	phys. on	Δ
uniform	scratch	0.192	0.750	**+0.558**
	fine-tune	0.131	0.166	+0.035
parabola (r/m-OOD)	scratch	0.343	0.375	+0.03
	fine-tune	0.124	0.198	+0.07
collision	scratch	0.538	0.635	+0.097
	fine-tune	0.393	0.525	+0.132

Table 2: Injection on/off × training regime, per domain (latent nMSE ↓ on the hardest clean split; injected structure: strict $a{=}g$ head + fixed slot). *Scratch* trains everything from random initialization; *fine-tune* starts the encoder from pretrained weights. Δ is the cost of switching injection on within a regime: positive in all six regime–domain cells, and amplified from +0.035 to +0.558 on uniform when physics is baked in from the start.

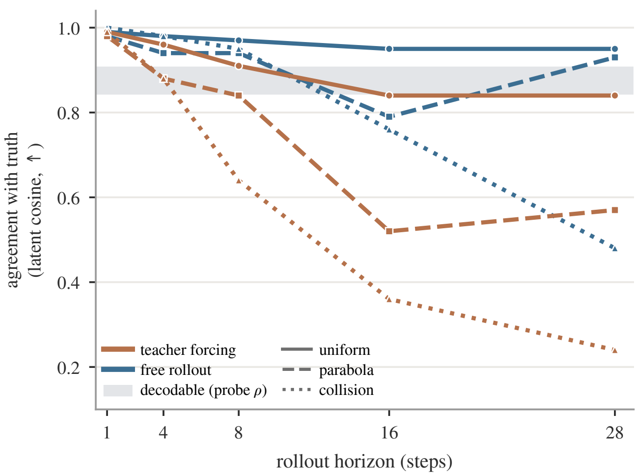

Figure 3: Decodable, but not load-bearing, in all three domains. Colour is the training protocol; line style is the domain. The shaded band is position decodability from the frame embeddings (both-OOD probe ρ, mean over both coordinates): the state is *present* to the same degree everywhere (0.84–0.91) and does not decay with horizon. The teacher-forced rollout nevertheless falls out of that band by an amount that tracks dynamics complexity—at horizon 28 it reaches 0.84 on uniform motion, 0.57 on parabola and 0.24 on collision, from 0.98–0.99 at one step. Free rollout (§3.4) recovers much of the gap with no physics added.

(fixed slot, consistency, consistency+acceleration) degrades per-scenario rollout nMSE by 3–10× against the identically trained free-rollout model (Appendix B); the gaps dwarf the three-seed noise band of the baseline. The probe variants have not yet been run on Physion++, so this corroboration covers the slot and consistency mechanisms only.

4.3 Mechanism: Decodable but Not Load-Bearing

The puzzle of §1 localizes here: in all three domains, position is linearly decodable from the latent at every horizon, yet the rollout drifts away from it, and the further the harder the dynamics (Figure 3)—*decodable, but not load-bearing*. The same dissociation is known for language models—probe-recoverable information need not be information the model uses (Elazar et al. 2021); the three tests explain why it arises here.

	uniform	parabola	collision
The bypass: position decodability ρ (both-OOD)			
full 192-d latent	0.92	0.85	0.78
black box only (190 d)	0.92	0.85	0.79
black box, slot pinned	0.92	0.86	0.79
the pinned slot itself (2 d)	0.92	0.85	0.51
random-2-d control		0.2–0.5	
The cost: what injection does to the encoder			
loss ratio $(\lambda\,\mathcal{L}_{\text{phys}})/\mathcal{L}_{\text{pred}}$	44×	125×	15×
effective dim, $\lambda{=}1 \to 10$	$41 \to 4$	$13 \to 3$	$28 \to 17$
collapse	−90%	−75%	−39%

Table 3: Why injection cannot help: the black box already carries position, and forcing it elsewhere costs the encoder. *Top*—the 190 non-slot dimensions decode position as well as all 192, and pinning position into a slot does not diminish that copy, so the predictor always has a route around the slot; a random-2-d control fails, so the redundancy is distributed rather than sparse. On collision the pinned slot is the *only* degraded read (0.51), yet the black box is untouched (0.79)—the bypass at its clearest. *Bottom*—the constraint nevertheless dominates optimization (weighted physical loss 15–125× the prediction loss at $\lambda{=}10$, a gradient-magnitude proxy) and collapses the encoder's participation ratio by 39–90%. Acting strongly on a quantity the latent already encodes is exactly the failure mode.

Test 1: the physical constraint is injected. Probing *only the* 190 *non-slot dimensions* decodes position as well as the full latent in all three domains, undiminished by pinning the slot, while a random-2-dimension control fails (Table 3, top). Position is a redundant, distributed copy the predictor can route through while ignoring any structured slot.

Test 2: the constraint acts harmfully. Both measurements move, which indicates the constraint acts (Table 3, bottom): at probe weight $\lambda{=}10$ the weighted physical loss outweighs the prediction loss by one to two orders of magnitude (a gradient-magnitude proxy), and the encoder's effective dimensionality collapses with it. At $\lambda{=}1$, the setting of §4.2, the ratio is milder (3–20×) yet the harm persists. The constraint demonstrably reshapes the representation—in a direction that competes with prediction.

Test 3: re-weighting the physical loss does not help. Sweeping the slot weight $w{=}1 \to 300$ across every partition (Appendix B) moves the harm systematically—the mechanism's direction is confirmed—but weighting rescues only 2 of 4 domain–partition decision cells, to *parity* and never a net gain: uniform both-OOD returns exactly to baseline (0.132 ± 0.017 vs. 0.136 ± 0.008), uniform r/m-OOD never recovers, and collision worsens monotonically with weight. Insufficient physical loss is therefore not the root cause: the

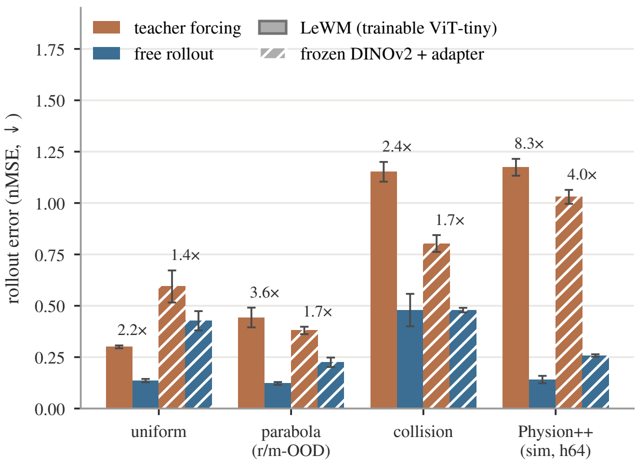

Figure 4: Teacher forcing vs. free rollout, both backbones on one axis (hardest clean split, nMSE ↓; solid = LeWM, hatched = the frozen-DINOv2 variant of §3.5; fold-change above each pair). Bars are three-seed means, error bars one sample std over seeds $3072/1234/42$: every teacher-forcing/free-rollout interval is disjoint. The physics-free protocol switch is worth 2.2–$8.3\times$ on LeWM and 1.4–$4.0\times$ on frozen DINOv2—the lever is backbone-independent, and largest where the dynamics are hardest (collision, Physion++).

black-box route stays open at every weight, and closing it requires changing the architecture.

4.4 Free Rollout Reduces Error by 2.2–$8.3\times$

Replacing teacher forcing with free rollout cuts the hardest-split error 2.2–$3.6\times$ across the three domains and $8.3\times$ on Physion++, every teacher-forcing/free-rollout interval disjoint over three seeds, and the whole pattern replicates on the frozen-DINOv2 backbone (Figure 4; per-domain values in Appendix C). Crucially the gain is not an OOD patch: *every* partition improves, including ID (2.0–$4.6\times$). The horizon must match dynamics complexity: collision improves markedly at $H{\approx}20$ ($0.393 \rightarrow 0.294$) while smooth domains degrade with excess horizon (uniform horizon-28 cosine $0.969 \rightarrow 0.814$ at $H{=}16$); realistic-simulation scenes have the largest appetite—extending $H{=}8 \rightarrow 28$ (with geometric scale jitter stabilizing amplitude; Appendix D) drives horizon-64 nMSE from 0.280 to 0.014, a $\sim$19$\times$ reduction with no inflection found (Appendix D).

As the positive control of §3.4, this result carries the methodological weight of the paper: on the same code, metrics, seeds, and data that judged all thirty injection cells, a physics-free intervention produces multiples—so the injection failure of §4.2 is attributable to the injected structure, not to the pipeline or the yardstick.

4.5 External Control: Extrinsic Is Necessary, Not Sufficient

Extrinsic port. The faithful extrinsic port (PIWM; §3.5) learns the correct physics and, where its encoder is in-distribution, beats our shared-latent model: rolled-out position ρ reaches 0.96 (ID) and 0.97 (v-OOD) vs. LeWM's $0.93/0.87$. But under radius/mass shift—which changes appearance, hitting the VAE encoder—it collapses to $\rho = 0.33$ (r/m-OOD) and 0.48 (both-OOD) while LeWM holds $0.89/0.87$; appearance-changing shift is an axis its original case studies do not vary. The architectural prediction of §4.3 is confirmed from the outside: making the physical state load-bearing by construction is what extrinsic designs buy (their equations alone do not transfer, §4.2); yet closing the bypass leaves the encoder-OOD problem unsolved. Extrinsic structure is necessary but not sufficient.

Cross-backbone replication. On the frozen-DINOv2 variant every headline finding replicates across three seeds. Presence is even stronger—the physics-naive frozen encoder decodes position at $\rho{=}0.95$ (vs. 0.90 for LeWM), with the same black-box bypass—so presence and bypass are properties of generic visual features, not our training. The protocol lever transfers (1.4–$1.7\times$ synthetic, $4.0\times$ Physion++; hatched bars in Figure 4). Injection still never wins ($27/30$ at or below baseline; Appendix C); the three above-parity cells dissolve under content-free controls—a shuffled target, or a $10\times$-weaker pin at matched loss weighting, keeps the apparent gain while ID error degrades, i.e. capacity-restriction regularization, not physics (Trap 5, Appendix E). One difference: with the encoder frozen, injection is inert rather than harmful—the losses reshape only the adapter, bounding damage but foreclosing benefit. Across every control, the same sentence survives: physical state in a latent world model is *decodable but not load-bearing*.

5 Conclusion

We asked whether injecting physics can make a latent world model physically consistent, and answered with a controlled anatomy: injection fails in 29 of 30 mechanism–domain cells—with the exact gravitational form, with perfect labels, and *more* when co-trained from scratch. The reason is that the state is already there: the non-slot dimensions carry the same position ($\rho = 0.79$–0.92), so an injected copy adds no signal while consuming capacity and gradient—*decodable but not load-bearing*. All three tests support this account, and both of its predictions held: re-weighting the slot reaches only parity, and an architecture that closes the bypass does help, until an OOD shift reaches its encoder. Closing the bypass is necessary, not sufficient. What moved prediction was never the structure but the training protocol, worth 2.2–$8.3\times$ on identical code and metrics.

The constructive reading: an inductive bias must supply what the latent is missing—*how the state evolves, not what the state is*. Two routes stay open: injections that target dynamics or conserved quantities rather than state, and extrinsic designs paired with OOD-robust encoders. Our own evolution-targeting variants are deliberately simple instantiations spanning the design axes; their failure shows that pinning the evolution is not *automatically* useful, not that a stronger, dynamics-matched design could not work.

Limitations. Our training-dynamics conclusions rest primarily on one compact backbone ($\sim$10M ViT-tiny JEPA);

the headline findings replicate on a frozen-DINOv2 backbone (§4.5), where injection is inert rather than harmful—so the *harm* claim is established on the trainable-encoder regime. Headline comparisons, re-weighting, and the seed-refuted cells are three-seed; the remaining 26 scan cells and the Physion++ variants are single-seed, with effects 5–20 seed-std and 3–10× clear of the noise band. The probe variants remain untested on Physion++, and the one favorable cell (velocity on parabola) suggests dynamics-matched injection deserves study we did not perform.

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A Training and Evaluation Details

Reproducibility. All training configurations, per-run logs, evaluation protocols, and the exact numbers behind every table and figure are archived alongside the code; the study builds on public assets (LeWM, PhyWorld, Physion, Physion++). We will release the evaluation suite, including the per-horizon/per-partition audit tools and the random-encoder controls.

Hyperparameters. All PhyWorld runs: AdamW, lr 5×10^{-5} (from-scratch reruns: 2×10^{-5}), weight decay 10^{-3}, bf16, gradient clip 1.0, 20 epochs (60 for from-scratch cells; the pretraining-regime 2×2 uses 60 epochs uniformly), batch size 64 for free rollout ($H{=}8$), 128 for teacher forcing ($H{=}1$), 48 for $H\geq16$. History size 3; embedding dimension 192; SIGReg weight 0.09. Seeds: $3072/1234/42$ where seed-replicated. Physion++ runs use the readout split (800 clips) with the same optimizer.

OOD partitions. ID: radius/mass $\in[0.7,1.5]$, velocity $\in[1.0,4.0]$. OOD draws radius/mass from $[0.3,0.6]\cup[1.5,2.0]$ and velocity from $[0,0.8]\cup[4.5,6.0]$. Collision uses the four-column variant (two objects' masses and velocities). Training sets are ID-only (1,000 trajectories); evaluation covers all four partitions ($\geq$1,600 trajectories per domain).

Probe protocol. Ridge regression ($\alpha{=}1$), trajectory-level 80/20 splits, probing the encoder output (not the projector), pretrained "paper" initialization, and $K{=}4$ stacked consecutive latents for velocity. Each choice matters: the naive combination (random init, single frame, through-projector) reads $\rho{=}0.166$ for uniform velocity where the corrected protocol reads 0.939 (Trap 4, Appendix E). Random-encoder and pixel-statistics controls are run alongside every probe.

B Full Injection Results

Table 4 is the numeric form of Figure 2, with the seed annotations spelled out.

Probe/slot factorial (uniform, free rollout). Latent both-OOD nMSE / pixel both-OOD PSNR (dB): baseline 0.131 / 20.41; probe-only 0.167 / 19.83; slot+reweight 0.114 / 21.30; probe+slot+reweight 0.125 / 20.02. The combination underperforms the slot alone on both scales; the latent and pixel verdicts agree.

Slot reweighting sweep (uniform). Slot weight 1/30/100: 0.183 / 0.114 / 0.136 (single seed), with the $w{=}30$ cell seed-replicated at 0.132 ± 0.017 against the baseline 0.136 ± 0.008. Kinematic head on the reweighted slot: velocity-OOD decoded position $\rho{=}0.991/0.987$ (x/y); uniform pixel horizon-28 23.34 vs. 22.09 dB (+1.25); appearance-OOD remains the weak axis (decoded ρ $0.29\rightarrow0.43$, vs. 0.96 without the kinematic head).

Bypass probe details. Protocol for §4.3, which reports the numbers. Two models are probed: the free-rollout baseline (no injection) and the slot-pinned model (structpos, $w{=}30$). For each, a ridge probe is fit on the dimension subset alone—the full 192, the black box [2:192], the slot [0:2], or a random pair as control—with train/test trajectories split 80/20, so decodability is measured out of sample. Values are Pearson ρ against ground-truth position on the both-OOD partition, averaged over both coordinates, from frame embeddings rather than rolled-out latents. The slot absorbs position (its supervision loss falls $2.53\rightarrow0.015$) without displacing the black box's redundant copy.

	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)
Free-rollout baseline	0.131	0.127	0.393
Fixed slot	0.183	0.156	0.651
+reweight ($w{=}30$)	0.114[s]	0.160	0.596
+velocity	0.207	**0.096**[m]	0.621
Probe	0.167	0.115[n]	0.647
+slot+reweight	0.125[s]	0.127	0.607
Kinematic (free MLP)	0.155	0.178	0.560
Kinematic (strict $a{=}g$)	0.206	0.173	0.559
Consistency	0.151	0.147	0.57–0.64
Label-free prior	0.171	0.172	0.653
Grounded prior	0.166	0.156	0.525

Table 4: The 30-cell scan (hardest clean split, latent nMSE $\downarrow$). [m]The one seed-robust gain: 0.096 ± 0.007 over three seeds vs. baseline 0.122 ± 0.007, non-overlapping. [n]Refuted by seeds: three-seed 0.137 ± 0.034 vs. baseline 0.122 ± 0.007—the single-seed low was seed luck. [s]Also refuted by seeds: reweighted slot 0.132 ± 0.017 and probe+slot 0.142 ± 0.017 vs. baseline 0.136 ± 0.008 (parity). Remaining cells are single-seed but sit 5–20 seed-std above baseline.

Encoded quantity vs. domain (position monotone, velocity sign-flipping). The reweighted slot's effect is systematic in what it encodes (hardest clean split; baseline: uniform both-OOD 0.131, parabola r/m-OOD 0.127, collision both-OOD 0.393):

slot content	uniform	parabola	collision
position (structpos)	0.132	0.160	0.596
position+velocity	0.207	0.096	0.621
Δ (adding velocity)	+0.075	−0.064	+0.025

Position is an output variable with the same role in every domain, so pinning it degrades monotonically with dynamics complexity and never flips sign. The marginal effect of adding velocity ($+0.075/-0.064/+0.025$) tracks velocity's role in each domain—redundant constant (uniform), linear driver (parabola), discontinuous jump (collision). The parabola gain ($0.122\pm0.007\rightarrow0.096\pm0.007$, every seed lower) is thus the sole seed-robust structural gain in the grid: a fixed form landing on a matching domain, not a portable prior.

Physion++ per-scenario (all ten scenarios, FR at 20 epochs). Rollout nMSE: bouncy_platform 0.041, bouncy_wall 0.094, friction_cloth 0.071, friction_collision 0.062, friction_platform 0.041, mass_collision 0.083, mass_dominoes 0.058, mass_waterpush 0.122; deformable scenes are the common hard case for every method (deform_clothhang 1.375, deform_clothhit 0.772). Injection

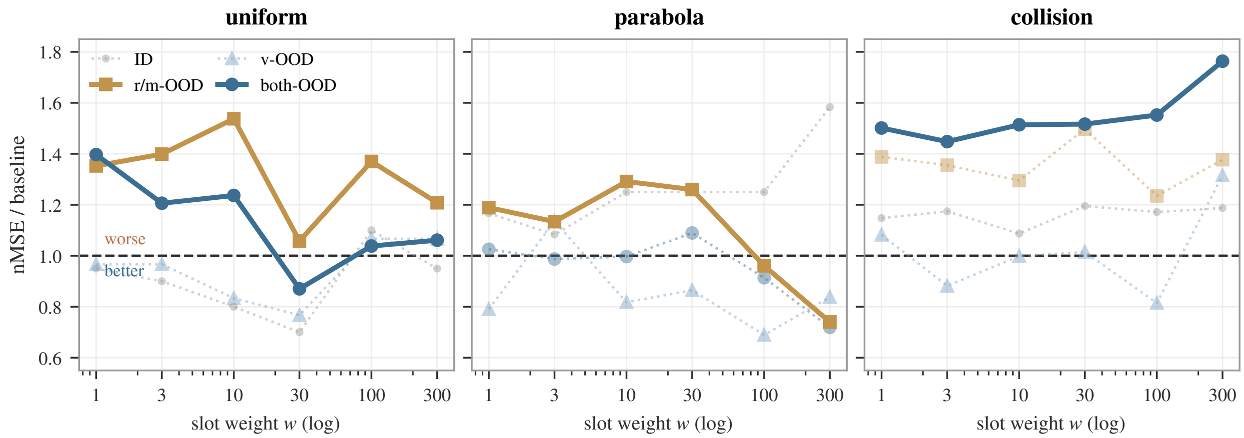

Figure 5: Dose–response re-weighting, all partitions. Solid curves are the four decision cells (hardest clean split); dotted are partitions we do not judge on—ID is not an extrapolation test, and parabola's v-OOD/both-OOD nMSE is denominator-degenerate (Appendix E). Only 2 of the 4 decision cells reach parity (uniform both-OOD, parabola r/m-OOD); uniform r/m-OOD and collision both-OOD stay above baseline at every weight, and collision degrades as weight grows. The sweep is single-seed; both parity cells were re-run over three seeds and land at baseline (§4.3).

Scenario	FR	+slot	+cons.	+cons.+acc.
friction_platform	**0.041**	0.119	0.078	0.067
mass_collision	**0.083**	0.510	0.682	0.696
mass_dominoes	**0.058**	0.452	0.586	0.597
mass_waterpush	**0.122**	0.363	0.663	1.476

Table 5: Physion++ direct training: per-scenario rollout nMSE (↓) at 20 epochs. Injection degrades realistic-simulation prediction by 3–10× on the mass/friction scenarios shown (deformable-object scenes are the hardest category for all methods, including the baseline). Single-seed; the gaps far exceed baseline seed noise.

variants degrade the rigid-body scenarios by 3–10× (Table 5) while leaving the deformable failure in place.

Physion++ horizons. Overall nMSE at horizons 16/32/64: baseline (H=8) 0.016/0.140/0.280; H=20 0.022/0.033/0.220; H=16+scale 0.010/0.035/0.136; H=20+scale 0.012/0.024/0.087; H=28+scale reaches horizon-64 nMSE 0.014. Free rollout is the main long-horizon lever (horizon-32 cosine 0.91 vs. 0.79–0.87 for every physics-augmented variant); geometric scale jitter acts as an amplitude stabilizer for long rollouts: without it, H=20 explodes on friction-collision (nMSE 24.6 at cosine 0.99); with it, 0.019.

C Cross-Backbone Replication: Frozen DINOv2

The cross-backbone variant (§3.5) freezes a DINOv2-small encoder (Oquab et al. 2024)—in the style of DINO-WM (Zhou et al. 2025)—and trains only the projector (as adapter) and the identical predictor stack, losses, and seeds (∼130 runs). Free-rollout baselines (three-seed): uniform both-OOD 0.427±0.039, parabola r/m-OOD 0.225±0.019, collision both-OOD 0.479±0.009; Physion++ horizon-64 0.258±0.005 vs. teacher forcing 1.030±0.028.

Teacher forcing vs. free rollout, per domain. Values behind Figure 4 (hardest clean split, latent nMSE ↓, three-seed mean±sample std). LeWM: uniform 0.300±0.007 → 0.136±0.008 (2.2×), parabola 0.443±0.048 → 0.122±0.007 (3.6×), collision 1.152±0.048 → 0.479±0.079 (2.4×), Physion++ horizon-64 1.174±0.041 → 0.141±0.018 (8.3×). Every teacher-forcing interval is disjoint from its free-rollout counterpart. The frozen-DINOv2 counterparts above give 1.4–4.0× over the same four settings, so the lever survives replacing a trainable ViT-tiny with a frozen general-purpose encoder.

Presence and bypass. The frozen encoder, which has never seen PhyWorld and received no physics supervision or gradient, decodes position from real-frame embeddings at ρ=0.951 (both-OOD)—*higher* than the PhyWorld-trained LeWM encoder (0.899). The 190 non-slot dimensions alone decode at 0.951, indistinguishable from the full latent; pinning the slot at weight 300 leaves the black-box copy at 0.973. Presence and the bypass are properties of generic visual representations.

Inert, not harmful. Unlike the trainable-encoder regime, the core variants (re-weighted slot, probe, consistency) land within the baseline's seed band on all three domains, while the plain slot (1.42× on uniform) and the dynamics-head variants (1.15–1.29× on all domains) still hurt. With the encoder frozen, injection losses can only reshape the 384→192 adapter: the bypass cannot be displaced, which bounds the damage—and forecloses the benefit. Injection-variant variance also roughly doubles (±0.087 vs. baseline ±0.039 on uniform), so single-seed cells on this backbone are uninformative.

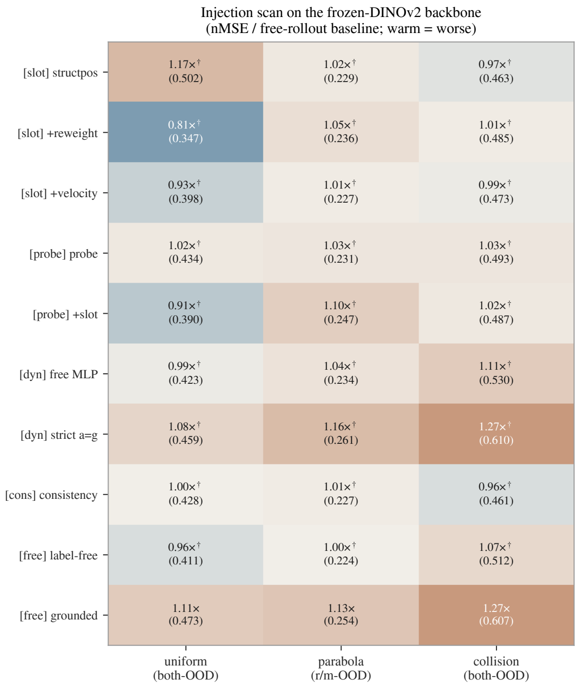

Figure 6: The injection scan repeated on the frozen-DINOv2 backbone (nMSE over the free-rollout baseline; mean over available seeds, † = three seeds). 10 cells worse / 17 parity / 3 better (27/30 at or below baseline). The dynamics-head variants are the worst here too (strict $a{=}g$ and the grounded prior reach $1.27\times$ on collision); the three above-parity cells, all on uniform, are the re-weighting confound of Trap 5.

D Augmentation: a Domain-Specific Lever with a Reversal Boundary

Augmentation is not part of the paper's main line (it is neither an injection mechanism nor a protocol variable), but its boundary behavior is a useful caution. On synthetic domains, per-trial appearance jitter (brightness/contrast, strength 0.5) roughly halves the hardest split: uniform $0.131 \to 0.068$ (-48%), parabola -63% (r/m-OOD $0.127 \to 0.065$); geometric scale jitter, only in combination with a long horizon, sets the collision record ($0.393 \to 0.172 \pm 0.004$ over three seeds). Interactions are non-monotone: appearance conflicts with long-horizon training (0.472, worse than either alone), scale synergizes (0.208), the triple is worse than the best pair (0.253); temporal-stride augmentation loses even to appearance on velocity-OOD (0.556 vs. 0.376), leaving unseen velocities the open hard case.

The appearance recipe reverses on realistic simulation: the same jitter that halves synthetic OOD error degrades Physion++ friction-collision prediction by two orders of magnitude (nMSE $0.062 \to 6.44$) while the *aggregate* long-horizon cosine improves (Trap 1 below), and it does not help zero-shot transfer either (0.597, below the 0.607 random-prior ceiling). The boundary is principled: appearance jitter encodes an appearance-invariance assumption that holds under PhyWorld's simple renderer but breaks where appearance carries physical information (friction, mass, material). Scale jitter, which assumes only size-invariance, keeps helping at realistic-simulation scale (Appendix B); appearance jitter flips sign. Augmentation gains are domain- and type-specific, not portable.

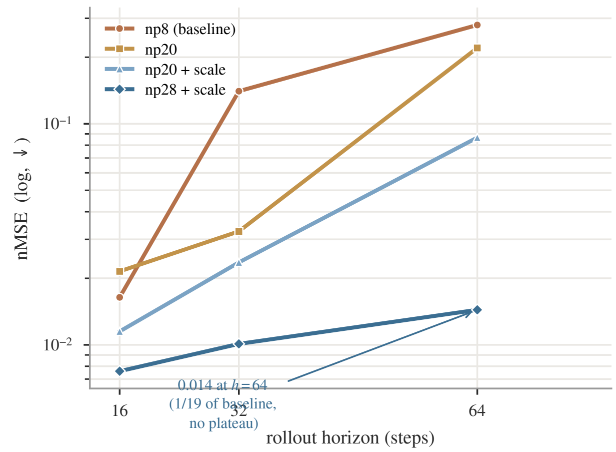

Figure 7: Physion++ direct training: per-horizon rollout nMSE (log scale) as the training horizon grows ($H{=}8 \to 28$, +scale jitter). Long-horizon error falls monotonically to $1/19$ of baseline with no inflection: realistic-simulation dynamics reward long-rollout training more than the synthetic domains do.

E Metric Pathologies and Evaluation Traps

Our anatomy repeatedly reached conclusions opposite to those suggested by common metrics; the decision rules of §4.1 distill the following five failure modes, each of which reversed at least one finding.

Trap 1: cosine and probe metrics are duals of the training loss. Probe correlation and latent cosine rise mechanically when probe or slot losses are optimized—they measure whether information is *present*, not whether prediction *uses* it. Probe supervision lifts velocity decodability from $\rho{=}0.44$ to 0.80 and gives the most stable long-horizon *cosine* (0.882 vs. 0.843), while its *pixel* rollout is the worst in the comparison (19.93 vs. 20.64 dB at horizon 28). Appearance augmentation on Physion++ improves the aggregate long-horizon cosine while per-scene nMSE collapses up to $100\times$ (deform_clothhit: cosine $0.610 \to 0.913$, nMSE $0.772 \to 3.69$). Judging either result by direction-only or readout metrics inverts the conclusion. Our own early sweep was misled this way: judged on $K{=}4$ probe ρ, $\lambda{=}50$ "won"; re-judged on prediction loss, the ranking reversed.

Trap 2: nMSE denominators can degenerate. When the target sequence degenerates (a parabola ball exits the frame, target variance $\to 0$), per-horizon nMSE explodes—3×10^4–2×10^6 at horizon $\sim$28 across all six parabola baseline variants, while cosine sits at a healthy 0.55–0.95—and pollutes every aggregate that includes late horizons. This is why parabola is judged on its clean r/m-OOD partition throughout. Traps 1 and 2 point in opposite directions (cosine under-reports failure; degenerate nMSE over-reports it); neither metric family is safe alone, hence per-partition, per-horizon cross-validation.

Trap 3: zero-shot transfer is bounded by the architecture prior. On Physion OCP, a *randomly initialized* encoder scores mean AUC 0.607. No synthetic-training configuration exceeds it: free rollout approaches it (0.603), physics variants fall below it (0.551–0.582), appearance augmentation 0.579–0.597; training longer approaches the ceiling monotonically (0.554 at epoch 1 to 0.603 at epoch 20) without crossing. Per-scenario, only Support ($+0.10$) and Collide ($+0.07$) show signal above the random control. Zero-shot claims must be calibrated against the random-prior floor, or they measure the architecture, not the learning.

Trap 4: protocol confounds fabricate results in both directions. Three probe-protocol choices (random vs. pretrained initialization, single-frame vs. stacked probes, probing through the projector) multiply into a false negative: the naive stack reads uniform velocity decodability at $R^2{=}0.166$, the corrected protocol 0.939. A silent initialization bug (only 24 of 216 loadable encoder keys actually loaded) invalidated an entire 45-configuration sweep before a loading guard caught it. And converged training loss proves nothing about rollout: our clean from-scratch baseline converges to the pretrained model's prediction loss (0.006 vs. 0.005) while its rollout OOD error is $2.8\times$ worse.

Trap 5: loss re-weighting masquerades as physics. On the frozen-DINOv2 backbone, heavily re-weighting the pinned position slot (weight 300; the 2 slot dimensions then carry 76% of the prediction loss) improves uniform both-OOD from $0.427{\pm}0.039$ to $0.286{\pm}0.015$—apparently the one clear injection win on either backbone. Two content-free controls dissolve it: pinning the slot to a *shuffled*, physically meaningless target at the same weight reaches $0.330{\pm}0.013$, and weakening the pin $10\times$ while keeping the weighting reaches $0.329{\pm}0.084$—most of the gain, with no usable physics. All three variants degrade in-distribution error ($0.099 \to 0.115$–0.136), the signature of capacity restriction (the predictor is effectively reduced to a 2-dimensional task), and the same recipe *worsens* collision ($1.11\times$) and parabola ($1.14\times$). An apparent injection gain must be checked against a content-free control at matched loss weighting.

Auxiliary-loss dominance (measurement behind §4.3, Test 2). At probe weight $\lambda{=}10$, the weighted auxiliary-to-prediction loss ratios—used as a proxy for gradient magnitude; we do not measure gradient norms directly—are $15\times$ (collision), $44\times$ (uniform), and $125\times$ (parabola); the encoder's effective (participation-ratio) dimensionality collapses by 39–90% relative to baseline (uniform $41 \to 4$). At $\lambda{=}1$ the ratios are benign, which is the setting used in §4.2.

Checklist. (i) Report per-partition *and* per-horizon; never aggregate across horizons without inspecting them. (ii) Judge with scale-aware prediction metrics (nMSE, decoded-pixel PSNR); use cosine and probe ρ as diagnostics only. (iii) Audit nMSE denominators for degenerate targets. (iv) Calibrate transfer against a random-init control, and validate probe protocols on a random encoder.